DD-Mamba: A Dual-Domain State-Space Forecasting Framework with Linear Anchors and Zero-Initialized Corrections

Anonymous Submission

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Abstract. Long-term multivariate time series forecasting must reconcile two complementary views: local, trend-driven dynamics in the time domain, and global periodic structure most compact in the frequency domain. Existing state-space (Mamba) forecasters model one view but not both, and deep forecasters often discard simple linear maps that remain near state-of-the-art. We present DD-Mamba, a dual-domain forecasting *framework* whose time branch (decomposition with a selective state-space encoder) and frequency branch (learned spectral filtering) each produce a forecast, fused by a convex gate that preserves both prediction paths. Two principles organize the framework: (i) *linear anchoring* — the time branch carries a DLinear-style linear pathway, the frequency branch an optional FITS-style spectral map, and all nonlinear heads are zero-initialized, so at initialization the model reduces to a linear map of the instance-normalized window; and (ii) *capacity placement* — a bidirectional Mamba over the *variate* dimension carries the model on high-channel data and is disabled on few-channel data where it overfits. On nine benchmarks under the look-back-96 protocol (three seeds), DD-Mamba reports the lowest mean MSE on seven, improving over reported S-Mamba results on five (up to 17% on Solar), tying three within noise, and trailing one, with fewer parameters on small datasets. Paired three-seed ablations on three datasets validate capacity placement in both directions (the variate mixer helps weather yet hurts ETTh1; normalization helps weather yet hurt solar), show a significant frequency-branch contribution on Solar with redundancy elsewhere, and find zero initialization accuracy-neutral; dual-domain modeling is a per-dataset structural option, not a universal gain.

Keywords: Time series forecasting · State-space models · Mamba · Frequency domain · Multivariate analysis

1 Introduction

Long-term forecasting of multivariate time series underpins applications from power-grid load planning to traffic management and weather-dependent operations. The field has cycled through architectural paradigms at a remarkable pace: Transformer variants with efficient attention [27, 21, 28], patch-based attention

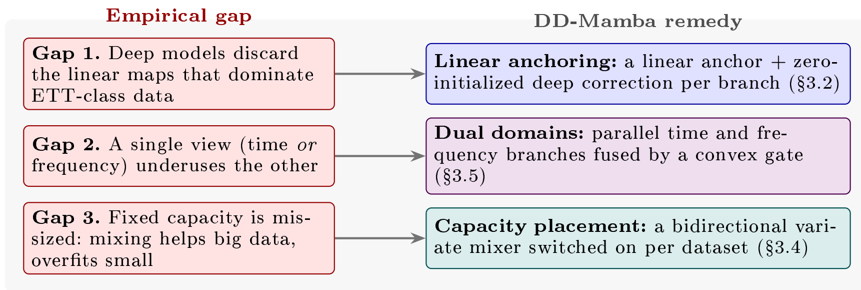

Fig. 1. The three empirical gaps that motivate DD-Mamba and the remedy each maps to, colour-matched to Fig. 2: time/linear (blue), dual-domain fusion (violet), and capacity placement (teal).

[17], inverted variate-token attention [14], and, most recently, selective state-space models (SSMs) built on Mamba [9, 20], which offer linear-time sequence modeling with input-dependent dynamics.

Three empirical gaps complicate this progress. **Gap 1: deep models discard strong linear maps.** A per-component linear projection from the look-back window to the horizon (DLinear [25]) and a tiny complex-linear interpolation of the low-passed spectrum (FITS [23]) match or beat far larger models on several benchmarks, especially the ETT family; deep architectures that ignore this spend their early training re-discovering a linear map, and on small datasets often never do. **Gap 2: the useful inductive bias differs by domain.** Local trends and regime changes are naturally expressed in the time domain, while multi-scale seasonality is compact in the frequency domain, yet models committed to a single view — attention or SSMs over time steps only, or spectral mixing only [24] — systematically underuse the other. **Gap 3: the right amount of capacity is dataset-dependent.** Cross-variate mixing is the decisive component on high-dimensional benchmarks [14, 20] yet pure overfitting capacity on low-channel data, so any *fixed* architecture is mis-sized for one regime or the other.

We answer these three gaps with DD-Mamba (Fig. 2), mapping each to one structural remedy (Fig. 1): *(i)* an explicit linear anchor with zero-initialized non-linear corrections in each branch; *(ii)* parallel time and frequency branches, fused by a convex gate that preserves both prediction paths; and *(iii)* a bidirectional variate mixer switched on or off per dataset. Each remedy is detailed below.

Dual branches with structurally preserved paths. Each branch outputs a complete forecast and the prediction is their learned *convex combination*, gated on both branches' features; being elementwise between the two, both domains keep an explicit path, and the gate can saturate near 0 or 1. Our paired ablations find the frequency branch significant on one of three datasets (Solar) and within noise elsewhere (Sect. 4.3) — per-dataset flexibility, not a universal dual-domain gain.

Linear anchors with zero-initialized corrections. Each branch carries an explicit *linear* pathway — DLinear-style trend/seasonal projections [25] in time, an optional FITS-style spectral map [23] in frequency — trained jointly. With all nonlinear heads zero-initialized and the gate opening at g_0, the model at initialization reduces to a linear map of the instance-normalized window (Sect. 3.2) — the RLinear-class structure [11, 10]; a controlled test finds this accuracy-neutral, so we position it as an interpretability default.

Mamba where it pays. A causal Mamba models the time axis; a *bidirectional* Mamba optionally handles spectral encoding and cross-variate mixing (S-Mamba style [20]). Capacity is *placed* per dataset, and the ablations support this both ways: removing the variate mixer hurts weather while *adding* it hurts ETTh1, each with a CI excluding zero (Sect. 4.3).

Contributions. (1) A dual-domain architecture whose output is a gated convex combination of per-branch forecasts, exposing each branch's per-sample weight; **(2)** a linear-anchoring scheme under which the model reduces to a linear map of the normalized window at initialization, shown accuracy-neutral in a controlled test; **(3)** a capacity-placement recipe for selective SSMs — causal Mamba over time, bidirectional Mamba over variates, neither when data is scarce; and **(4)** the lowest reported average MSE on seven of nine benchmarks under the look-back-96 protocol (three-seed mean$\pm$std), with a released harness and paired multi-seed ablations including negative, dataset-dependent findings.

2 Related Work

Transformer forecasters. Informer [27], Autoformer [21] and FEDformer [28] adapt attention to long sequences via sparsity, decomposition, and frequency mixing; Non-stationary Transformers [15] restore the stationarity that normalization removes, and Crossformer [26] attends across both time and variates. PatchTST [17] treats sub-series patches as tokens with channel independence; iTransformer [14] inverts the axes and attends over *variates*, establishing that cross-channel modeling drives accuracy on high-dimensional benchmarks. DD-Mamba adopts the variate-mixing insight with linear-time bidirectional SSMs, coupled to an explicit frequency branch and linear anchors.

Linear and frequency-domain models. DLinear/NLinear [25] showed decomposition plus a linear map rivals Transformers; RLinear [11] added reversible instance normalization [10], and the MLP-based TiDE [6], TSMixer [3], TimeMixer [19] and deep-basis N-HiTS [2] confirm that lightweight mixers stay competitive. In the spectrum, FreTS [24] learns frequency-domain MLPs and FITS [23] forecasts with a single complex-linear interpolation of the low-passed spectrum at negligible cost. DD-Mamba does not compete with these models — it *contains* them: each branch carries a linear pathway of its domain's family, with the deep machinery trained as a correction.

State-space models. Mamba [9] introduced selective SSMs with input-dependent dynamics and linear-time scans, extended by Mamba-2 [5]. In forecasting, S-Mamba [20] and TimeMachine [1] apply bidirectional Mamba over variate tokens, Bi-Mamba+ [12] adds forget-gated bidirectional blocks, and further works explore frequency-domain SSMs [16]. DD-Mamba unifies these threads: causal Mamba over time, bidirectional Mamba over frequency bins and variates, placed per dataset rather than uniformly.

Decomposition and heteroscedasticity. Splitting a series into interpretable components is a century-old idea: classical additive models, STL [4], wavelet analysis, EMD, and SSA extract a trend, a seasonal part, and an irregular remainder, and learned decompositions such as SCINet [13] and MICN [18] carry it into deep forecasters. A recurring limitation, made explicit by the seasonal-trend-*dispersion* decomposition (STD) [7], is that these methods model *level* but not *scale*: when a series' variance changes over time — heteroscedasticity — the fluctuation is pushed into the irregular term and left unmodeled, exactly the regime of the non-stationary forecasting benchmarks. DD-Mamba inherits this lineage but gives each classical component a learnable home. Its time branch does moving-average trend/seasonal decomposition in the spirit of DLinear [25], so the seasonal-trend split lands on the *linear anchors* (Sect. 3.2); its RevIN instance normalization [10] rescales each window by its own mean and standard deviation — a *local scale-normalization related to*, but not equivalent to, STD's dispersion component. The connection is motivation, not identity: RevIN normalizes rather than decomposes, does not extract dispersion as a separate interpretable series, and is reversible, so we do not claim it implements an STD-style decomposition. What it does do is remove per-window scale before the anchors act, which is why our ablations find normalization decisive yet dataset-dependent (Sect. 4.3), helping where windows carry genuine scale shift and hurting where apparent nonstationarity is itself periodic structure. The discipline of routing a specific deficiency to a dedicated spectral or normalization operator rather than applying one mechanism uniformly has also been argued for frequency-domain design in adjacent fields [8].

3 Method

3.1 Problem Formulation and Normalization

Given a look-back window $\mathbf{X} \in \mathbb{R}^{L \times C}$ of L steps and C variates, the task is to predict the next H steps $\mathbf{Y} \in \mathbb{R}^{H \times C}$. Following RevIN [10], we standardize each window per variate, $\tilde{\mathbf{X}} = (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$, and de-normalize the final forecast. Instance normalization is treated as a per-dataset switch rather than a universal default; Sect. 4.3 shows why.

3.2 Time Branch: Decomposition, Linear Backbone, Selective Scan

The time branch decomposes $\tilde{\mathbf{X}}$ with a moving average into trend $\mathbf{X}_t$ and seasonal $\mathbf{X}_s$ components [21, 25] and forecasts with two cooperating paths.

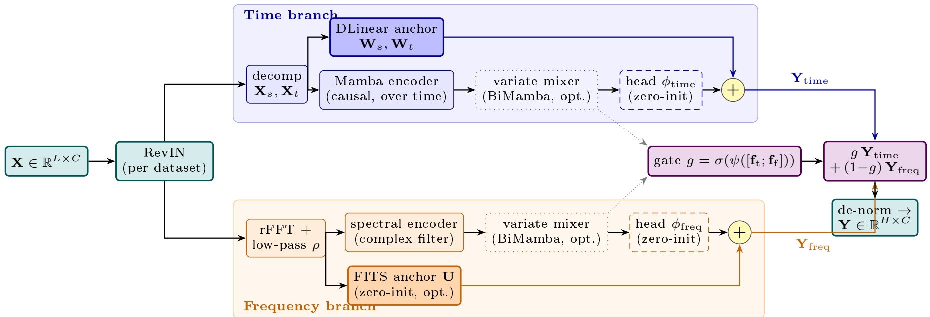

Fig. 2. DD-Mamba overview, color-coded by role: the **normalization** wrapper (teal), the **time branch** (blue), the **frequency branch** (orange), and the **convex-gate fusion** (violet). In each branch a solid-filled *linear anchor* runs in parallel with a zero-initialized nonlinear head (dashed), so at initialization the model reduces to a linear map of the instance-normalized window. Dotted boxes are per-dataset switches; dotted grey arrows carry branch features to the gate, whose convex weight g mixes the two coloured forecast paths $\mathbf{Y}_{\text{time}}$ and $\mathbf{Y}_{\text{freq}}$.

Linear backbone. Per-component linear maps project the window directly to the horizon:

$$\mathbf{Y}_{\text{lin}} = \mathbf{W}_s\mathbf{X}_s + \mathbf{W}_t\mathbf{X}_t \in \mathbb{R}^{H\times C}, \qquad \mathbf{W}_s, \mathbf{W}_t \in \mathbb{R}^{H\times L}, \tag{1}$$

i.e., a DLinear-style map [25] (randomly initialized and trained jointly — a parameterized pathway, not a fitted DLinear), shared across variates.

Selective-scan correction. The pair $[\mathbf{X}_s; \mathbf{X}_t]$ is embedded per time step and processed channel-independently by a stack of Mamba layers [9] (Fig. 3a). A Mamba layer discretizes a state-space system with input-dependent parameters $(\boldsymbol{\Delta}, \mathbf{B}, \mathbf{C})$:

$$\mathbf{h}_k = e^{\boldsymbol{\Delta}_k \mathbf{A}}\,\mathbf{h}_{k-1} + \boldsymbol{\Delta}_k \mathbf{B}_k\, u_k, \qquad y_k = \mathbf{C}_k^{\top}\mathbf{h}_k + D\, u_k, \tag{2}$$

scanned causally over the L steps and applied independently to each of the d_{in} feature dimensions: per dimension, the state is $\mathbf{h}_k \in \mathbb{R}^N$, $\mathbf{A} \in \mathbb{R}^N$ is diagonal, the input u_k is scalar, and $(\boldsymbol{\Delta}_k \in \mathbb{R},\ \mathbf{B}_k, \mathbf{C}_k \in \mathbb{R}^N)$ are produced from u by learned linear maps [9]. The last state, summed with a whole-window linear embedding $\mathbf{W}_e\tilde{\mathbf{x}}^{(c)}$ (one per variate, as in [20]), forms the branch feature $\mathbf{f}_{\text{time}} \in \mathbb{R}^{C\times d}$. A head ϕ_{time} maps features to a horizon correction:

$$\mathbf{Y}_{\text{time}} = \mathbf{Y}_{\text{lin}} + \phi_{\text{time}}(\mathbf{f}_{\text{time}}), \qquad \phi_{\text{time}} \equiv \mathbf{0} \text{ at init.} \tag{3}$$

Because ϕ_{time} is zero-initialized, the branch's output at initialization equals its (randomly initialized) DLinear-style backbone; the SSM learns only what the linear map cannot express. We run the recurrence in fp32 even under mixed precision, as the L-step accumulation overflows in half precision and silently destroys training.

3.3 Frequency Branch: Spectral Filtering and a FITS Anchor

The frequency branch applies the real FFT along time, $\mathbf{Z} = \mathrm{rFFT}(\tilde{\mathbf{X}}) \in \mathbb{C}^{C\times F}$, $F = \lfloor L/2 \rfloor + 1$, optionally low-passed by zeroing the top fraction ρ of bins.

Spectral encoder. A learnable complex linear filter $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i \in \mathbb{C}^{F\times F}$ densely mixes bins, $\mathbf{Z}' = \mathbf{Z}\mathbf{W}^\top$ along the frequency axis (a bidirectional Mamba over the bin sequence is an input-dependent alternative). The (real, imaginary) parts are projected to the branch feature $\mathbf{f}_{\text{freq}} \in \mathbb{R}^{C\times d}$ and a head ϕ_{freq} produces the spectral forecast.

FITS anchor (optional, per dataset). Mirroring the time branch, a complex linear map $\mathbf{U} \in \mathbb{C}^{F'\times F}$ with $F' = \lfloor (L+H)/2 \rfloor + 1$, applied along the frequency axis ($\mathbf{Z}\mathbf{U}^\top \in \mathbb{C}^{C\times F'}$), maps the (low-passed) window spectrum to that of a length-$(L+H)$ series, whose inverse transform's last H samples are the branch's linear spectral forecast:

$$\mathbf{Y}_{\text{freq}} = \underbrace{\mathrm{irFFT}_{L+H}(\mathbf{Z}\mathbf{U}^\top)[\,L{:}\,]}_{\text{linear-in-frequency}} + \phi_{\text{freq}}(\mathbf{f}_{\text{freq}}), \qquad \mathbf{U}, \phi_{\text{freq}} \equiv \mathbf{0} \text{ at init.} \tag{4}$$

This pathway *parameterizes* the FITS [23] family, but unlike FITS itself $\mathbf{U}$ is *zero-initialized*, so the frequency branch is silent at initialization. This is deliberate: the frequency grids of a length-L and a length-$(L+H)$ window do not align (harmonic k maps to the non-integer index $k(L+H)/L$), so no canonical identity initialization exists and a random $\mathbf{U}$ would inject noise during the highest-learning-rate epochs. With the time head, spectral map, and gate all zero or constant at init, the end-to-end map in the instance-normalized space is the linear $g_0 \cdot \mathbf{Y}_{\text{lin}}$; composed with RevIN, whose statistics are themselves nonlinear in the input, this is a randomly initialized linear map wrapped in instance normalization — exactly the RLinear-class structure [11, 10], linear in the normalized window (not the raw input) and by construction (not fitted). The anchor is enabled per dataset (Table 2); where disabled, the frequency branch stays active through its encoder and head but carries no linear pathway, so linear anchoring then rests on the time branch alone.

3.4 Cross-Channel Mixing over Variates

Channel independence is a strong baseline [17, 25] but leaves accuracy on the table when variates are coupled [14, 20]. Inside each branch, before its head, we optionally run M layers of *bidirectional* Mamba over the C per-channel features, followed by a position-wise feed-forward block (Fig. 3b). Bidirectionality is essential: channel order carries no causal direction, so each layer sums a forward and a reversed scan in one residual stream. The mixer is a per-dataset choice: $M{=}2$ at width 512 on electricity/traffic (its single most load-bearing component; Sect. 4.3), $M{=}2$ at width 256 on weather, $M{=}1$ on solar, and $M{=}0$ on the few-channel ETT and exchange datasets, where it triples parameters without adding signal.

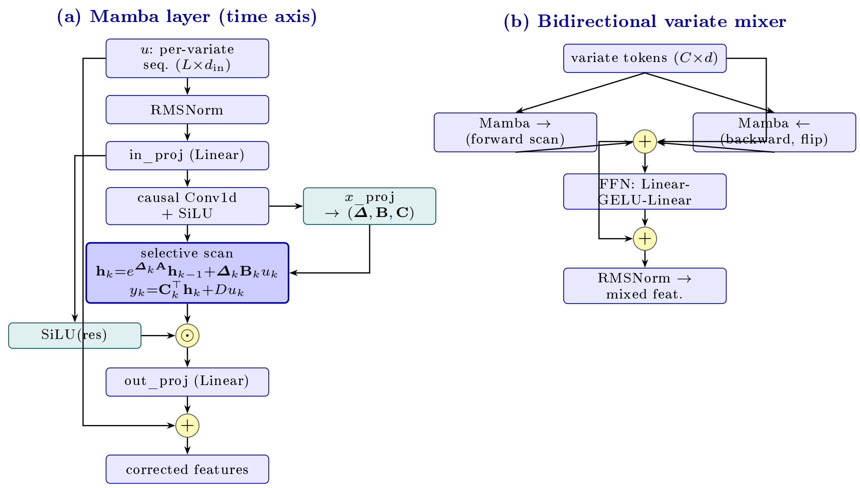

Fig. 3. Internal structure of DD-Mamba's two selective-state-space blocks (cf. the blue boxes in Fig. 2). **(a)** The causal Mamba layer used as the time-axis encoder (Eq. (2)): an input-dependent selective scan, gated by a SiLU branch and wrapped in a residual, with $(\boldsymbol{\Delta}, \mathbf{B}, \mathbf{C})$ predicted per step from the input. **(b)** The bidirectional mixer used over the *variate* dimension (Sect. 3.4): forward and backward scans of the C channel tokens summed in one residual stream, then a position-wise feed-forward block (the S-Mamba recipe [20]).

3.5 Forecast-Level Convex Fusion

Each branch emits a forecast and a feature. The model output is a gated convex combination

$$\mathbf{Y} = g \odot \mathbf{Y}_{\text{time}} + (1-g) \odot \mathbf{Y}_{\text{freq}}, \qquad g = \sigma\big(\psi([\mathbf{f}_{\text{time}}; \mathbf{f}_{\text{freq}}])\big) \in (0,1)^{C\times 1}, \quad (5)$$

with ψ applied row-wise (per variate), followed by RevIN de-normalization. Since $\mathbf{Y}$ lies elementwise between the branch forecasts, both domains retain an explicit prediction path, and the gate value reports each branch's convex mixing weight — a lightweight proxy for its contribution, not a causal attribution. The constraint is structural, not behavioral: g is a sigmoid and can saturate near 0 or 1, so one branch may contribute very little — by design, as the ablations show the useful domain is dataset-dependent. ψ is zero-initialized with bias 2.2, so training starts at $g \approx 0.9$: the time branch is linear at initialization while the frequency branch is silent, and an even 0.5/0.5 mix would waste the highest-learning-rate epochs on noise.

3.6 Training

We minimize MSE with AdamW under a halving schedule ($\eta_e = \eta_0 \cdot 2^{-(e-1)}$), which concentrates learning in the first epochs and suppresses the late-epoch

Table 1. Benchmark datasets and where DD-Mamba places its capacity.

Dataset	Variates	Steps	Granularity	Variate mixer	Params
ETTh1/ETTh2	7	17,420	1 hour	off	0.25M
ETTm1/ETTm2	7	69,680	15 min	off	0.24–0.36M
Weather	21	52,696	10 min	M=2, d=256	5.8M
Solar-Energy	137	52,560	10 min	M=1, d=128	0.96M
Electricity	321	26,304	1 hour	M=2, d=512	19.7M
Traffic	862	17,544	1 hour	M=2, d=512	19.7M
Exchange	8	7,588	1 day	off	0.09M

overfitting that cosine schedules invite on small benchmarks; all runs use 10 epochs with best-checkpoint early stopping.

4 Experiments

4.1 Setup

Datasets and protocol. We evaluate on nine standard benchmarks (Table 1) under the unified protocol of [14, 20]: look-back 96, horizons 96/192/336/720. ETT uses the canonical 12/4/4-month border split, the others chronological 0.7/0.1/0.2 splits; variates are standardized by training-set statistics; metrics are MSE and MAE.

Baselines. We compare against the strongest published results under this protocol: S-Mamba [20], iTransformer [14], PatchTST [17], DLinear [25], and TimesNet [22]. Numbers are quoted from [20] rather than reproduced: the protocol is identical, but training pipelines, seeds, and implementations differ across works, so sub-noise-floor gaps to quoted numbers should be read with caution (flagged in the text); a unified rerun is planned for the camera-ready appendix.

Implementation and Reproducibility. All experiments use PyTorch 2.11 (CUDA 13.0) on a single NVIDIA RTX 5090. Optimization is AdamW (grad clip 5.0), 10 epochs with the learning rate halved each epoch, early stopping on validation loss, and mixed precision with the selective scan forced to fp32 (Sect. 3.2). The decomposition kernel is 25; Mamba blocks use expansion 2 and conv width 4 throughout. Table 2 lists all per-dataset settings; configs and the ablation harness ship with the code. All main results are means over *three seeds* (2024–2026); the maximum per-entry std across the 36 dataset–horizon entries is 0.005 MSE, typically 0.001–0.003.

Statistical methodology. We compare at two levels. *Within our framework* (ablations), every variant shares seeds with the reference, so we report *paired* per-seed differences with t-based 95% CIs (n=3) and treat a component as supported

Table 2. Per-dataset hyperparameters. d: model width; time enc.: time-branch encoder (Mamba layers in parentheses); M: variate-mixer layers; N: SSM state size; ρ: spectral low-pass fraction (Sect. 3.3); FITS: the *optional* spectral linear anchor of Eq. (4) — when "no", the frequency branch remains fully active (rFFT, spectral encoder, forecast head) but carries no explicit linear pathway; IN: instance normalization (RevIN); do.: dropout; B: batch size; wd: weight decay. †Solar adopts IN = no from the ablation finding (Sect. 4.3); its main results were re-measured under this configuration across all horizons and seeds.

Dataset	d	time enc.	M	N	ρ	FITS	IN	lr	B	do./wd
ETTh1	128	Mamba (1)	0	16	0.4	yes	yes	$5{\cdot}10^{-4}$	32	$0.2/10^{-4}$
ETTh2	128	Mamba (1)	0	16	0.3	yes	yes	$5{\cdot}10^{-4}$	32	$0.3/5{\cdot}10^{-4}$
ETTm1	128	Mamba (2)	0	16	0	no	yes	10^{-4}	32	$0.1/10^{-4}$
ETTm2	128	Mamba (1)	0	16	0.2	no	yes	10^{-4}	32	$0.2/2{\cdot}10^{-4}$
Weather	256	Mamba (2)	2	16	0	yes	yes	$2{\cdot}10^{-4}$	32	$0.1/10^{-4}$
Solar	128	Mamba (2)	1	16	0	no	no†	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Electricity	512	MLP	2	32	0	no	yes	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Traffic	512	MLP	2	32	0	no	yes	10^{-3}	16	$0.1/10^{-4}$
Exchange	64	Mamba (1)	0	8	0.5	no	yes	10^{-4}	32	$0.3/5{\cdot}10^{-4}$

Table 3. Long-term forecasting results averaged over horizons $H \in \{96, 192, 336, 720\}$ with look-back L=96 (MSE/MAE; lower is better). DD-Mamba entries are means over three seeds (Table 4 reports per-horizon std). Baselines quoted from [20]. **Bold** marks the best result *and* every result within the run-to-run noise floor (< 0.005, Sect. 4.1) of it, treated as tied best; <u>underline</u> marks the next-best result.

Dataset	DD-Mamba (ours)	S-Mamba	iTransformer	PatchTST	DLinear	TimesNet
ETTh1	**0.439**/**0.434**	0.455/0.450	<u>0.454</u>/<u>0.447</u>	0.469/0.454	0.456/0.452	0.458/0.450
ETTh2	**0.374**/**0.401**	<u>0.381</u>/**0.405**	0.383/<u>0.407</u>	0.387/<u>0.407</u>	0.559/0.515	0.414/0.427
ETTm1	**0.386**/**0.398**	<u>0.398</u>/<u>0.405</u>	0.407/0.410	**0.387**/**0.400**	0.403/0.407	0.400/0.406
ETTm2	**0.280**/**0.325**	<u>0.288</u>/<u>0.332</u>	<u>0.288</u>/<u>0.332</u>	**0.281**/**0.326**	0.350/0.401	0.291/0.333
Weather	**0.248**/**0.276**	**0.251**/**0.276**	<u>0.258</u>/**0.278**	0.259/<u>0.281</u>	0.265/0.317	0.259/0.287
Electricity	**0.168**/**0.263**	**0.170**/**0.265**	<u>0.178</u>/<u>0.270</u>	0.205/0.290	0.212/0.300	0.192/0.295
Traffic	0.438/<u>0.281</u>	**0.414**/**0.276**	<u>0.428</u>/0.282	0.481/0.304	0.625/0.383	0.620/0.336
Exchange	0.363/**0.405**	0.367/<u>0.408</u>	<u>0.360</u>/**0.403**	0.367/**0.404**	**0.354**/0.414	0.416/0.443
Solar	**0.199**/**0.260**	0.240/<u>0.273</u>	<u>0.233</u>/**0.262**	0.270/0.307	0.330/0.401	0.301/0.319

only when its CI excludes zero. *Against quoted baselines*, pairing is impossible, so we use a conservative unpaired tie threshold of 0.005 MSE. Our in-framework DLinear re-run shows why: through the identical pipeline on ETTh1 it averages 0.462/0.456 MSE/MAE versus its quoted 0.456/0.452 — a +0.006 implementation gap, the order of several margins in Table 3. Cross-paper comparisons therefore cannot support sub-0.01 distinctions; DD-Mamba's margin over the in-framework DLinear (0.439 vs. 0.462) is, by contrast, a same-pipeline comparison.

4.2 Main Results

Under the noise-floor tie rule, DD-Mamba reports the lowest average MSE on seven of nine datasets (Table 3; per-horizon values with standard deviations in Table 4). The margin is clear on Solar (0.199 vs. iTransformer's 0.233; −17%

Table 4. DD-Mamba per-horizon results (mean ± std over three seeds), look-back L=96.

Dataset		H=96	H=192	H=336	H=720	Avg
ETTh1	MSE	0.380±.001	0.431±.002	0.474±.004	0.470±.003	0.439
	MAE	0.396±.001	0.424±.001	0.447±.003	0.468±.001	0.434
ETTh2	MSE	0.293±.001	0.368±.003	0.414±.001	0.421±.001	0.374
	MAE	0.344±.002	0.391±.000	0.426±.000	0.441±.001	0.401
ETTm1	MSE	0.320±.001	0.366±.002	0.398±.001	0.462±.001	0.386
	MAE	0.359±.001	0.384±.001	0.406±.001	0.443±.001	0.398
ETTm2	MSE	0.177±.000	0.242±.002	0.302±.000	0.401±.003	0.280
	MAE	0.260±.001	0.303±.001	0.340±.000	0.398±.002	0.325
Weather	MSE	0.162±.000	0.212±.000	0.269±.001	0.352±.003	0.248
	MAE	0.207±.001	0.255±.000	0.295±.002	0.349±.002	0.276
Electricity	MSE	0.140±.000	0.157±.001	0.174±.001	0.200±.004	0.168
	MAE	0.235±.000	0.252±.001	0.271±.001	0.296±.004	0.263
Traffic	MSE	0.402±.002	0.424±.001	0.441±.001	0.485±.003	0.438
	MAE	0.266±.001	0.275±.001	0.283±.000	0.301±.000	0.281
Exchange	MSE	0.086±.000	0.179±.001	0.330±.002	0.857±.001	0.363
	MAE	0.205±.000	0.301±.000	0.416±.002	0.699±.000	0.405
Solar	MSE	0.180±.004	0.195±.005	0.207±.001	0.212±.005	0.199
	MAE	0.237±.003	0.259±.003	0.270±.001	0.274±.005	0.260

vs. S-Mamba, with the instance-normalization-free configuration of Sect. 4.3), ETTh1 (−0.015), and ETTh2 (−0.007); on ETTm1/ETTm2 it beats S-Mamba clearly but ties PatchTST (0.001), and on Weather/Electricity the lead is below the floor (tied). DD-Mamba stays competitive on Exchange — a near random-walk where all methods cluster within 0.013 MSE and DLinear leads by 0.009 — and trails S-Mamba on Traffic by +0.024 MSE, which we *hypothesize* reflects the weekly cycle exceeding the look-back (Sect. 5). As all comparisons are to numbers *reported* in [20] under the same protocol, we phrase the claim conservatively: DD-Mamba is competitive with results reported under this protocol, with its largest margins on Solar (normalization) and the small ETT datasets, where the linear anchors and capacity restraint matter most and DD-Mamba uses ≈ 0.25M parameters — two orders of magnitude fewer than variate-attention models.

4.3 Component Ablations

Our ablation harness flips exactly one switch of a dataset's full configuration, holding split, schedule, seeds, and every other hyperparameter fixed. We report its complete output on three datasets under their *final released configurations*, three seeds each at H=96: ETTh1 and Solar (Table 5) and Weather (Table 6), with the paired methodology of Sect. 4.1. Two consistency checks pass: the Solar "full" row reproduces the main-table Solar H=96 entry (0.180±.004), and its w/o-instance-norm variant coincides with the full (RevIN-off) configuration, so that cell is omitted.

Dual-domain evidence. The decisive result is on Solar: removing the frequency branch degrades MSE by +0.0114 with a CI excluding zero — the first statistically supported frequency-branch contribution in our studies. On ETTh1 and

Table 5. Component ablations on ETTh1 and Solar-Energy (H=96, three seeds, final configurations): paired per-seed ΔMSE to the full model with t-based 95% CIs; * CI excludes zero. The FITS anchor is enabled on ETTh1 (hence *removed*) and disabled on Solar (hence *added*); the variate mixer is off on ETTh1 (*added*) and on for Solar (*removed*).

Variant	ETTh1: Δ [95% CI]	Solar: Δ [95% CI]
full (reference MSE)	0.3802$\pm$.001	0.1802$\pm$.004
time branch only (freq removed)	+0.0034 [−0.009, +0.016]	+0.0114 [+0.001, +0.021]*
freq branch only (time removed)	−0.0007 [−0.004, +0.003]	+0.0051 [−0.013, +0.023]
sum fusion	−0.0026 [−0.007, +0.002]	+0.0066 [−0.005, +0.018]
w/o instance norm	+0.0061 [−0.001, +0.013]	—
w/o linear backbone	−0.0031 [−0.009, +0.003]	+0.0081 [−0.019, +0.035]
random init (vs. zero-init)	−0.0012 [−0.005, +0.002]	−0.0046 [−0.022, +0.013]
FITS anchor removed / added	+0.0042 [−0.018, +0.026]	+0.0088 [−0.002, +0.019]
variate mixer added / removed	+0.0077 [+0.004, +0.012]*	+0.0119 [−0.010, +0.034]
Mamba→MLP over time	−0.0022 [−0.007, +0.003]	+0.0089 [−0.002, +0.020]
freq Mamba encoder	−0.0024 [−0.011, +0.006]	−0.0117 [−0.044, +0.020]

Weather both single-branch variants lie within their CIs, and on ETTh1 the frequency branch *alone* even matches the full model (−0.0007): the two branches are largely redundant there. The supported dual-domain claim is thus conditional — the frequency path contributes measurably on Solar and is redundant, but not harmful, elsewhere.

Capacity placement, validated in both directions. Adding the variate mixer on ETTh1 *hurts* (+0.0077*) while removing it on Weather also hurts (+0.0104*) — significant paired evidence on both sides of the decision. Normalization shows the same two-sided pattern: removing it hurts Weather (+0.0161*), while *disabling* it on Solar improved the re-measured main results from 0.228 to 0.199 average MSE.

Zero initialization: a null result. The controlled random-initialization comparison finds no significant accuracy difference on either dataset (−0.0012 on ETTh1, −0.0046 on Solar; both CIs include zero). We therefore make no accuracy claim for it: its value here is the interpretable linear start of Sect. 3.2, with stability analyses (convergence, failure rates) left to future work.

Weaker components. The DLinear backbone is not individually significant in any study, and the learned gate versus plain averaging is within CIs everywhere. On Weather (Table 6) exactly three components survive the paired test — normalization (+0.0161), the variate mixer (+0.0104), and the temporal Mamba (+0.0027, positive on all three seeds) — every other CI includes zero. The recurring pattern is that component value is dataset-dependent, in *both* directions for the mixer and normalization: the argument for shipping components as per-dataset switches, not universal defaults.

Normalization is a per-dataset decision. The probe that showed RevIN *hurting* solar (−0.016 MSE when removed) shows the opposite on traffic (removing it degrades 0.405→0.505 at H=96). The datasets look alike — bounded and strongly

Table 6. Component ablation on Weather (H=96; MSE mean $\pm$ std over three seeds). Δ is the *paired* per-seed difference to the full model with a t-based 95% confidence interval (n=3); * CI excludes zero.

Variant	MSE	paired Δ [95% CI]	Component isolated
full	0.1620$\pm$.0003	—	reference
w/o instance norm	0.1781$\pm$.0035	+0.0161 [+0.0053, +0.0269]*	RevIN
w/o channel mixer	0.1724$\pm$.0021	+0.0104 [+0.0037, +0.0172]*	cross-variate Mamba mixing
Mamba→MLP over time	0.1647$\pm$.0005	+0.0027 [+0.0020, +0.0034]*	selective scan (time)
w/o linear backbone	0.1631$\pm$.0006	+0.0011 [−0.0014, +0.0036]	DLinear anchor (time)
time branch only	0.1628$\pm$.0011	+0.0007 [−0.0020, +0.0035]	frequency branch removed
w/o FITS anchor	0.1627$\pm$.0016	+0.0007 [−0.0041, +0.0055]	spectral linear anchor
freq branch only	0.1626$\pm$.0014	+0.0006 [−0.0040, +0.0052]	time branch removed
freq Mamba encoder	0.1626$\pm$.0016	+0.0006 [−0.0044, +0.0055]	spectral encoder type
sum fusion	0.1622$\pm$.0007	+0.0001 [−0.0020, +0.0023]	learned gate → average

periodic — but solar's night zeros are *periodic structure* (window statistics then encode the night fraction, and de-normalization amplifies errors by an unstable std), whereas traffic windows carry genuine distribution shift (weekday/weekend, per-sensor levels) that normalization absorbs. We thus ship normalization as a verified per-dataset switch and caution against transferring its conclusions.

Negative results that shaped the design. Deepening the traffic variate mixer from two to four layers (19.7 → 37.9M parameters) gave no accuracy gain, so the release keeps two layers. Enabling the mixer on ETT (7 channels, 8.4K training windows) instead tripled parameters and *caused* premature early stopping through immediate overfitting; the released ETT configurations disable it.

4.4 Analysis

Where each domain wins. On solar the branch-only ablations (Table 5, Fig. 4b) show both domains contribute but the time view carries more (removing it costs +0.012 vs. +0.002 MSE) — consistent with the look-back hiding periodic structure (the 10-minute daily cycle spans 144>L=96 samples; likewise weather's daily and traffic's weekly cycles). The fusion gate (Eq. (5)) also exposes per-sample convex weights (a proxy, not a causal measure): Fig. 4a reads them on a trained ETTh1 model, where the gate settles below equal mix toward the frequency branch the ablation certifies as sufficient. Gate and ablation agree, though only the ablation is causal.

Efficiency profile. Table 7 reports three hardware-independent measurements per dataset: parameters, MACs per forecast, and the forward activation footprint. Parameter count is governed by model *width*, not channel count (electricity and traffic share 19.7M despite a 2.7× gap in variates), so model size is decoupled from data dimensionality. Compute and activation memory instead scale with the channel count through the variate mixer, which scans C tokens: traffic (862 channels) costs 60 GMACs per forecast against electricity's 22 at equal width, while ETT and exchange sit two orders of magnitude below (0.02–0.08 GMACs,

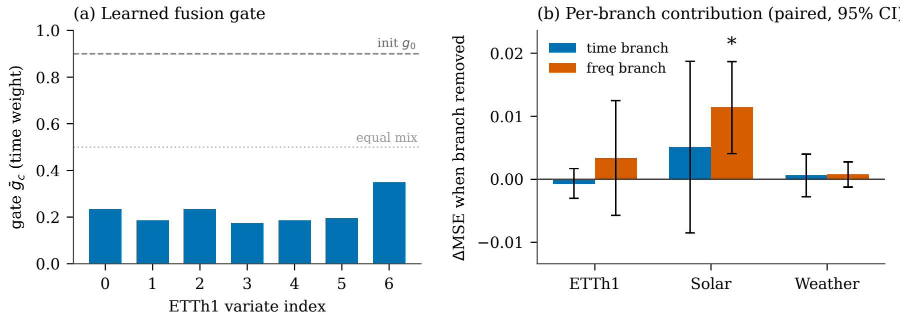

Fig. 4. Dual-domain evidence. **(a)** The learned fusion gate on ETTh1: the mean convex weight $\bar{g}_c$ assigned to the *time* branch for each variate over the test set. Initialized time-heavy at $g_0{\approx}0.9$ (dashed), the trained gate redistributes weight toward the frequency branch (below the equal-mix line, dotted), matching the ETTh1 ablation in which the frequency branch alone reproduces the full model. **(b)** Per-branch contribution: the paired increase in MSE (H=96, three seeds) when each branch is removed — freq contribution = time-only − full, time contribution = freq-only − full — with t-based 95% CIs; * marks a CI clearing zero. Only Solar's frequency branch is individually significant; on ETTh1 and Weather the branches are mutually redundant. This is the visual form of our conditional dual-domain claim: the two domains are complementary on some datasets and interchangeable on others.

<5 MB), making the capacity restraint there cheap as well as accurate. Wall-clock throughput and peak GPU memory under identical hardware are left for the extended version.

5 Conclusion

DD-Mamba is a dual-domain state-space forecasting *framework*: each branch carries a linear pathway of its domain, deep components are zero-initialized corrections, and capacity (especially cross-variate mixing) is placed per dataset. Under the look-back-96 protocol it achieves the lowest reported average MSE on seven of nine benchmarks and stays competitive on Exchange, while trailing S-Mamba on Traffic, at 0.09–19.7M parameters. We state plainly what the paired ablations support: capacity placement is validated in *both* directions (the variate mixer helps weather yet hurts ETTh1; normalization helps weather yet hurt solar), the temporal encoder helps weather, and the frequency branch contributes measurably on Solar while being redundant elsewhere; zero initialization is accuracy-neutral. We therefore claim per-dataset structural flexibility with dataset-dependent branch value, not a universal gain from combining domains — plus a transferable caution: normalization can help or hurt depending on whether apparent nonstationarity is genuine shift or periodic structure.

Limitations remain, each with the experiment that would resolve it. (1) The frequency-branch gain is shown on one dataset, one horizon, n=3; extending the

Table 7. Engineering profile of DD-Mamba per dataset (look-back 96, horizon 96, batch 1). GMACs: multiply–accumulates per forecast (`Linear` + `Conv1d` + the selective-scan recurrence); act. mem.: forward activation footprint (fp32). All three columns are hardware-independent; wall-clock throughput on the RTX 5090 is deferred to the extended version.

Dataset	Variates	Params (M)	GMACs	Act. mem. (MB)
Exchange	8	0.09	0.02	2.3
ETTh1	7	0.25	0.08	4.0
Weather	21	5.77	1.90	46.6
Solar	137	0.96	3.26	150.4
Electricity	321	19.70	22.4	254.3
Traffic	862	19.70	60.2	682.7

paired matrix would map where it generalizes. (2) Zero initialization is accuracy-neutral, but its hypothesized stability value is untested. (3) Baselines are quoted from [20], not re-run; the +0.006 in-framework DLinear gap (Sect. 4.1) means comparative claims hold only as "competitive with reported results" until the baselines are reproduced here. (4) The fixed look-back of 96 hides cycles longer than the window (weather's daily, traffic's weekly), our hypothesis for the traffic deficit. Multi-resolution windows and adaptive spectral cutoffs are natural next steps.

Reproducibility. Code, configurations, and the ablation harness are supplementary material, to be released publicly upon acceptance.

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